

IE 509: Lab 3 Assignment (Due 28th August)

1. (a) Write a python function *quadraticFunction(a,b,c)*, which return the quadratic function $f(x) = ax^2 + bx + c$, where input values a, b, c are the coefficients of the function $f(x)$.
(b) Write a python function *derivativeFunction(a,b,c)*, which return the derivative of the quadratic function.
(c) Write a function *rootFinder(a,b,c)* which return the real roots and complex roots of the quadratic function. where inputs a, b, c are the coefficients of the quadratic function $f(x) = ax^2 + bx + c$
2. (a) Consider a dictionary where the keys are labels and the values are numerical data. For example:

```
data = {'apple': 70, 'banana': 40, 'cherry': 20, 'date': 100, 'fig': 20, 'elderberry': 80, 'grape': 35, 'honeydew': 40, 'kiwi': 100, 'lemon': 18, 'mango': 50, 'nectarine': 22, 'orange': 45, 'pear': 100, 'plum': 33}
```


Using linear search, find the minimum value and maximum value among these dictionary values and return the label for this maximum and minimum values.
3. You are given a 2D matrix where each row is sorted in ascending order and each column is also sorted in ascending order. Implement a binary search algorithm to determine if a target value exists within the matrix. If the target exists, return true; otherwise, return false.
4. You are required to find the square root of a given non-negative integer x using a binary search algorithm. The result should be the integer part of the square root (i.e., the floor value), and if x is a perfect square, return the exact integer square root.
5. Write a recursive function to find and print the n th Fibonacci number. The Fibonacci numbers are defined by $F_0 = 0$ and $F_1 = 1$, and

$$F_{n+1} = F_n + F_{n-1}$$

Your task is to implement a recursive function that takes an integer n and returns the n th Fibonacci number.